$x = [x \; y \; z]^T$

then $2x^2 - y^2 - z^2 = x^T M x = 0$ where

$$M = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

For $f(x) = x^T M x$ [M- symmetric]

$$\nabla f = (M + M^T)x = 2Mx.$$

Normal at any point is parallel to $Mx$.

Putting in point P:

$x_p = [1 \; 0 \; \sqrt{2}]^T$

$$Mx_p = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \sqrt{2} \end{bmatrix}$$

$$= [2 \quad 0 \; \sqrt{2}]^T$$

Finding norm: $\sqrt{V \cdot V^T}$

$\hookrightarrow \|v\| = \sqrt{6}$

$$n = \frac{v}{\|v\|} = \frac{[2, 0, \sqrt{2}]^T}{\sqrt{6}}$$

$$= \frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}$$ (Option D)